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AI-LAB

Project-Proposal

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“Topic: Habit Tracker”

Introduction:

Developing good habits is an important part of achieving success and building self-discipline. However, it is not always easy to remember what we do every day or how consistent we are. This project helps users easily track their daily habits, see their progress, and stay consistent over time.

The program focuses on simple logic building using basic Python concepts. It does not use advanced features, which makes it beginner-friendly. The project can also be improved in the future, such as by saving data in files or creating a graphical interface

Description:

The Habit Tracker allows users to record daily habits they want to improve, such as **Exercise, Reading, Studying, Sleeping Early, or Drinking Water**. Each day, the program asks the user if they completed each habit. Based on the daily inputs, the program calculates how consistent the user has been.

In this improved version, the project also includes new features like updating habits, deleting habits, checking daily summary, viewing best and weak habits, and tracking streaks. These features make the tracker more helpful and practical.

The main purpose of this project is to motivate users to follow their routines and understand where they need improvement by showing a clear percentage of completion for each habit.

Objective:

The main objective of this project is to:

- Help users develop self-discipline through daily habit tracking.
- Provide a simple and user-friendly system to monitor progress.
- Demonstrate how basic Python features can be combined to create a working mini-project.
- Encourage students to apply logic and problem-solving skills in programming.

Components Used:

This project uses **only the basic concepts of Python**, which makes it beginner-friendly and easy to understand:

1. **Lists** – To store habits and progress.
2. **Loops (for, while)** – To repeat actions and display menus.
3. **If-Else Statements** – To make decisions based on user input.
4. **Datetime Module** – To show the current day of the week.

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5. **Basic Input and Output** – For user interaction.
6. **Functions** – Used to organize tasks like adding, updating, deleting habits, summaries, and streak features.

Working of the Project:

1. **Start-the-Program:**

The program displays a simple menu with options:

The program displays a menu with options:

- Add a new habit
- View habits
- Update habit
- Delete habit
- Track today's habits
- Daily Summary
- Weekly Summary
- Recommendation Section (Best & Weak Habit)
- Streak System
- Exit

2. **Adding-Habits:**

The user can add multiple habits that they want to monitor. These are stored in a list in the program's memory.

3. **Updating-Habits :**

If the user wants to rename or modify a habit, they can select the habit from the list and update it easily.

4. **Deleting-Habits :**

Users can delete habits they no longer want to track.

5. **Tracking-Daily-Habits:**

Each day, the user marks “y” or “n” for each habit. The data is stored for summaries.

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6. **Daily-Summary:**

Shows the user's performance for the current day for example, habits completed and missed.

7. **Weekly-Summary:**

The program calculates completion percentage for each habit using all recorded days.

8. **Recommendation-Section:**

Based on weekly performance, the program shows:

- **Best Habit (highest consistency)**
- **Weak Habit (lowest consistency)**

9. **Streak-System :**

Tracks how many days in a row the user has completed a habit without missing it.

10. **Exit:**

The user can exit anytime.

Example Output (Sample Run):

===== Habit Tracker =====

1. Add Habit
2. View Habits
3. Update Habit
4. Delete Habit
5. Track Today's Habits
6. Daily Summary
7. Weekly Summary
8. Recommendation Section
9. Streak System
10. Exit

Enter a habit to track: Exercise
Habit added successfully!

Choose an option (1-10): 1
Enter a habit to track: Study
Habit added successfully!

Choose an option (1-10): 3
Select habit number to update: 2
Enter new name for the habit: Reading
Habit updated successfully!

Choose an option (1-10): 4

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Select habit number to delete: 1

Habit deleted successfully!

Choose an option (1-10): 5

Tracking for: Monday

Did you complete 'Reading' today? (y/n): y

Today's progress saved!

Choose an option (1-10): 6

Daily Summary:

Completed Habits: 1/1

Completion Percentage: 100%

Choose an option (1-10): 7

Weekly Summary:

Reading: 100.0% consistency

Choose an option (1-10): 8

Recommendations:

Best Habit: Reading (100%)

Weak Habit: None

Choose an option (1-10): 9

Streaks:

Reading: 1 day streak

Choose an option (1-10): 10

Goodbye! Stay consistent!

Code:

```
import datetime
```

```
# Two lists to store data
```

```
habits = []          # Stores habit names
```

```
progress = []        # Stores daily progress
```

```
print("==== Habit Tracker =====")
```

```
# ===== 1. ADD A HABIT =====
```

```
def add_habit():
```

```
    name = input("Enter a new habit: ")
```

```
    if name != "":
```

```
        habits.append(name)
```

```
        print("Habit added!")
```

```
    else:
```

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===== 2. VIEW HABITS =====

```
def view_habits():
    if len(habits) == 0:
        print("No habits yet!")
        return

    print("Your Habits:")
    for i in range(len(habits)):
        print(f"{i+1}. {habits[i]}")
```

===== 3. UPDATE HABIT =====

```
def update_habit():
    if len(habits) == 0:
        print("No habits to update!")
        return

    view_habits()

    try:
        num = int(input("Which habit number do you want to update? "))
        if num < 1 or num > len(habits):
            print("Invalid number!")
            return
    except:
        print("Please enter a number!")
        return

    new_name = input("Enter new name: ")
    habits[num-1] = new_name
    print("Habit updated successfully!")
```

===== 4. DELETE HABIT =====

```
def delete_habit():
    if len(habits) == 0:
        print("No habits to delete!")
        return

    view_habits()

    try:
        num = int(input("Which habit number do you want to delete? "))
        if num < 1 or num > len(habits):
            print("Invalid number!")
            return
    except:
        print("Please enter a number!")
        return
```

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```
removed = habits.pop(num-1)
print(f'Habit '{removed}' deleted successfully!")
```

```
# ===== 5. TRACK TODAY =====
def track_today():
    if len(habits) == 0:
        print("Add habits first!")
        return

    today = datetime.date.today().strftime("%A")
    print(f"Today is: {today}")

    # Create list for today's progress
    today_progress = []

    # Ask about each habit
    for habit in habits:
        answer = input(f'Did you do '{habit}' today? (y/n): ").lower()
        if answer == "y":
            today_progress.append(1) # Done
        else:
            today_progress.append(0) # Not done

    # Save today's progress
    progress.append(today_progress)
    print("Today's progress saved!")

# ===== 6. DAILY SUMMARY =====
def daily_summary():
    if len(progress) == 0:
        print("No data yet!")
        return

    # Get most recent day
    today = progress[-1]

    # Count done habits
    done = 0
    for value in today:
        done += value

    total = len(today)
    percentage = (done / total) * 100

    print("Daily Summary:")
    print(f"Completed: {done}/{total}")
    print(f"Percentage: {percentage:.1f}%")
```

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===== 7. WEEKLY SUMMARY =====

```
def weekly_summary():
    # Check if we have any data
    if progress == []: # Same as len(progress) == 0
        print("No data yet! Track at least one day first.")
        return

    print("Weekly Summary:")

    # For each habit...
    for i in range(len(habits)):
        # Get habit name
        current_habit = habits[i]

        # Count how many times it was done
        times_done = 0
        for day_data in progress:
            if day_data[i] == 1:

times_done += 1

        # Calculate percentage
        total_tracked_days = len(progress)
        if total_tracked_days > 0:
            consistency = (times_done / total_tracked_days) * 100
        else:
            consistency = 0

        print(f'{current_habit}: {consistency:.1f}% consistent')
```

===== 8. RECOMMENDATIONS =====

```
def recommendations():
    if len(progress) == 0:
        print("No data yet!")
        return

    total_days = len(progress)
    percentages = []

    # Calculate percentage for each habit
    for i in range(len(habits)):
        done = 0
        for day in progress:
            done += day[i]
        percentages.append((done / total_days) * 100)

    # Find best and worst
```


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```
best = max(percentages)
weak = min(percentages)

print("Recommendations:")
print(f"Best Habit: {habits[percentages.index(best)]} ({best:.1f}%)")
if best == weak:
    print("All habits are equally consistent!")
else:
    print(f"Needs Improvement: {habits[percentages.index(weak)]} ({weak:.1f}%)")

# ===== 9. STREAK SYSTEM =====
def streak_system():
    # Check if we have any data
    if progress == []: # Same as len(progress) == 0
        print("No data yet! Track at least one day first.")
        return

    print("Current Streaks:")

    # For each habit...
    for i in range(len(habits)):
        # Get habit name
        current_habit = habits[i]

        # Check streak starting from today (most recent day)
        current_streak = 0

        # Start from the last day and go backwards
        day_index = len(progress) - 1 # Start with most recent day

        while day_index >= 0:
            # If habit was done on this day
            if progress[day_index][i] == 1:
                current_streak += 1
                day_index -= 1 # Move to previous day
            else:
                break # Stop if habit was missed

        print(f"{current_habit}: {current_streak} day streak")

# ===== MAIN MENU =====
while True:
    print("\n===== MAIN MENU =====")
    print("1. Add Habit")
    print("2. View Habits")
    print("3. Update Habit")
    print("4. Delete Habit")
    print("5. Track Today")
```

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```
print("6. Daily Summary")
print("7. Weekly Summary")
print("8. Recommendations")
print("9. Streaks")
print("10. Exit")

choice = input("Choose option (1-10): ")

if choice == "1":
    add_habit()
elif choice == "2":
    view_habits()
elif choice == "3":
    update_habit()
elif choice == "4":
    delete_habit()
elif choice == "5":
    track_today()
elif choice == "6":
    daily_summary()
elif choice == "7":
    weekly_summary()
elif choice == "8":
    recommendations()
elif choice == "9":
    streak_system()
elif choice == "10":
    print("Goodbye! Stay consistent!")
    break
else:
    print("Invalid choice! Try again.")
```

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Advantages:

- Encourages users to build and maintain good habits.
- Simple and clear design for beginners.
- Uses basic Python logic.
- Includes useful features like update, delete, summary, recommendations, and streak system.
- Can be expanded easily.
- Helps in personal productivity and growth.
-

Possible Future Improvements:

- Save progress permanently using files or databases.
- Add charts or graphs using libraries like matplotlib.
- Add notifications or reminders.
- Create a GUI version or mobile app.
- Improve the streak system with badges and rewards.

Conclusion:

The Habit Tracker is a beginner-friendly Python project that helps users stay consistent with their daily habits. It teaches important concepts like lists, loops, conditionals, and functions.

It is practical, simple, and gives students a clear example of how Python can be used to solve real-life problems and build discipline in both programming and personal life.



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